

Exercise and worked solution

Introduction to derivatives

Four parts to handle the building blocks of derivatives: a forward's linear payoff, an option's asymmetry, a speculator's leverage, and the mechanics of netting and margins. Work through each part before reading the solution.

Part A · Payoffs: forward vs option

PROBLEM

(a) A firm enters a long forward with delivery price $K = 1.10$ on 3M €. Payoff if $S_t = 1.15$? if $S_t = 1.06$? (b) A long call with strike 50 costs a premium of 4. Net payoff at $S_t = 60$ and $S_t = 45$? Break-even, max loss, max gain?

SOLUTION

- (a) Long payoff = $(S_t - K) \times 3M$. At 1.15: $(1.15 - 1.10) \times 3M = \mathbf{+150,000 \text{ USD}}$; at 1.06: $(1.06 - 1.10) \times 3M = \mathbf{-120,000 \text{ USD}}$. Linear and symmetric: the short counterparty gains/loses exactly the reverse. The forward is a firm commitment, one cannot 'not exercise'.
- (b) Net payoff = $\max(S_t - 50, 0) - 4$. At 60: $10 - 4 = \mathbf{+6}$; at 45: $0 - 4 = \mathbf{-4}$ (out-of-money option, loss of the premium). Break-even = $K + \text{premium} = \mathbf{54}$. Max loss = **the premium (4)**; max gain = **unlimited**. This is the asymmetry of buying an option: capped downside, open upside.

Part B · The speculator's leverage

PROBLEM

A stock trades at 40. A speculator has 4,000 USD. A call with strike $K = 42$ trades at 2. They buy either 100 shares or 2,000 calls. Profit if the stock ends at (a) 48, (b) 41? What does the leverage reveal?

SOLUTION

- (a) At 48: shares = $100 \times (48 - 40) = \mathbf{+800}$; calls = $2,000 \times \max(48 - 42, 0) - 4,000 = 2,000 \times 6 - 4,000 = 12,000 - 4,000 = \mathbf{+8,000}$.
- (b) At 41: shares = $100 \times (41 - 40) = \mathbf{+100}$; calls = $2,000 \times \max(41 - 42, 0) - 4,000 = 0 - 4,000 = \mathbf{-4,000}$ (out-of-money option, total loss of the premium).
- Leverage amplifies everything: the calls turn +800 into +8,000 when it works, but +100 into -4,000 when the stock stays below the strike. This is exactly the leverage that, diverted from hedging into hidden speculation, produced the Leeson, Kerviel, Adoboli disasters — hence the importance of internal controls.

Part C · The CCP and its four defences

PROBLEM

(a) A member holds 4 long and 1 short contracts on the same future: what is its net position after netting? (b) Does the variation margin earn interest? And the initial margin? Why the difference? (c) In what order does the waterfall absorb losses on a default?

SOLUTION

- (a) Net position = 4 longs - 1 short = **3 net longs**. Netting reduces the CCP's exposure to this member.
- (b) The variation margin earns **no** interest: it is a final daily settlement of the loss, not a deposit. The initial margin **earns** interest, as it remains the property of the member who posted it (securities accepted, with a haircut).
- (c) Waterfall: first the **defaulter's IM**, then its **default fund contribution**, then the **other members' contributions**, and as a last resort the **CCP's equity**. VM flow direction: price $\uparrow \rightarrow$ shorts pay longs; price $\downarrow \rightarrow$ the reverse.

Part D · Bilateral netting and central clearing

PROBLEM

Two firms A and B have four transactions worth, for A: +50, −20, +10, −30 M. (a) A's exposure if B goes bankrupt, without netting (cherry-picking)? (b) With a master agreement? (c) Does mandatory CCP clearing concern all derivatives?

SOLUTION

1. (a) Without netting, B's liquidator defaults on the contracts positive for A (+50, +10) and demands the negatives (−20, −30). A loses the sum of the positives: $50 + 10 = \mathbf{60M}$.
2. (b) With a master agreement, all is treated as a single net contract = $+50 - 20 + 10 - 30 = \mathbf{+10M}$: exposure falls to 10M. Netting works both ways (if A defaults, B loses nothing).
3. (c) **No**: mandatory CCP clearing (G-20 2009) targets only **standardised OTC derivatives between financial institutions**; ordinary end users are exempt. The CCP generally increases multilateral netting (15 → 6M in the GARP example) and mutualises losses, but creates moral hazard, adverse selection and procyclicality.

Key takeaways: forward = firm commitment, linear payoff $S_t - K$; option = right for a premium, capped downside and open upside (buying), unlimited loss for a call writer; leverage amplifies gains and losses; the CCP neutralises counterparty risk by novation + 4 defences (netting, VM no interest, IM with interest, default fund); the master agreement blocks cherry-picking; and mandatory CCP clearing concerns only standardised derivatives between financial institutions.